

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

*I _S.DIVYASRI(192524388)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: S.DIVYASRI

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

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Task 1: Article Summary

(a) Citation Details, Domain, and ML Problem Addressed

Full citation: Moslehi, S., Rabiei, N., Soltanian, A. R., & Mamani, M. (2022). Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran. *BMC Medical Informatics and Decision Making*, 22, Article 192. <https://doi.org/10.1186/s12911-022-01939-x>

Domain / application area: The article sits in the domain of clinical healthcare informatics, specifically interpretable machine-learning support for hospital mortality prognosis during an infectious-disease outbreak. It sits at the intersection of biostatistics and applied artificial intelligence, where the priority is not just predictive power but a model whose reasoning a physician can audit at the bedside.

Specific ML problem addressed: The authors treat COVID-19 patient outcome as a binary classification problem (dead vs. recovered) and build interpretable decision-tree models that not only predict the outcome but also expose which demographic, clinical, and laboratory factors drive that prediction, so that clinicians can use the resulting rules for rapid, explainable decision-making. This is a supervised-learning task in the strict sense: every training record already carries a known outcome label, and the objective is to induce a general decision rule that generalises correctly to new, unlabelled patients.

Why this article was selected: the paper was chosen because it covers three CO4 sub-topics in a single applied pipeline — Inductive Learning (rules learned purely from data), Decision Trees (four distinct tree variants benchmarked against one another), and Statistical Learning Methods (the LMT model embeds logistic regression at each node, and Random Forest is used as a statistical feature-importance ranking tool) — which makes it an unusually complete case study for this course outcome.

Article at a Glance

Authors	Samad Moslehi, Niloofar Rabiei, Ali Reza Soltanian, Mojgan Mamani
Year of Publication	2022 (published after 2019 — eligible per assignment guidelines)
Journal	BMC Medical Informatics and Decision Making (Springer Nature / BioMed Central)
Article Type	Original Research, Open Access (CC BY 4.0)
DOI	10.1186/s12911-022-01939-x
Study Design	Retrospective cohort study
Study Location	Sina and Besat Hospitals, Hamadan, western Iran
Study Period	February 2020 – July 2021
Sample Size	2,470 hospitalised COVID-19 patients (1,853 recovered; 617 deceased)
ML Techniques Used	Random Forest (feature selection); LMT, C4.5, C5.0, CART (classification)
Software / Tools	Python (scikit-learn) and R (train package), version 3.8 / 4.1.2
CO4 Topics Covered	Decision Trees, Inductive Learning, Statistical Learning Methods

(b) Summary of the Key Objective (150–200 words)

The central objective of the study is to identify the demographic, clinical, and laboratory features most strongly associated with COVID-19 mortality and to build an interpretable classification model capable of predicting that outcome with acceptable accuracy. The authors note that prior machine-learning studies on COVID-19 mortality largely reported classification accuracy figures without explaining which features drove the predictions or how those features interacted, which limited their usefulness for bedside decision-making. This is the research gap the paper sets out to close. To fill it, the authors first apply a random-forest procedure purely as a feature-ranking tool, then feed the top-ranked features into four transparent, rule-based decision-tree variants (LMT, C4.5, C5.0 and CART) rather than a black-box ensemble. The best-performing tree is presented as a readable flowchart with branch-level mortality-risk percentages, so that the model's classification logic — and not just its accuracy score — is directly interpretable by physicians making treatment-prioritisation decisions.

Framed against the wider literature, this objective matters because interpretability is often traded away for raw predictive accuracy in clinical machine learning; by benchmarking four transparent tree variants against each other rather than reaching for an opaque ensemble or deep model, the authors position interpretability itself as a primary design goal rather than an afterthought bolted onto a high-accuracy black box.

Task 2: Methodology Analysis

(a) ML Technique(s) Used and Dataset Description

The authors use a two-stage machine-learning pipeline. In the first stage, a Random Forest (an ensemble of 1,000 trees) is used purely for feature selection: the Gini-based relative importance of each candidate variable is computed, and only features above a data-driven cut-off (the third quartile, $\approx 6\%$) are retained. In the second stage, four individual decision-tree classifiers are trained on the retained features and compared: the Classification and Regression Tree (CART), C4.5, C5.0, and the Logistic Model Tree (LMT), which embeds a logistic-regression model at each tree node. The tree with the best combined accuracy/F1-score is retained as the final interpretable model, and its structure is visualised as a flowchart with per-branch mortality-risk estimates, supported by an ROC/AUC analysis.

Decision-Tree Variants Explained

Random Forest (pre-processing role)	An ensemble of 1,000 individual decision trees, each trained on a bootstrap sample of the data. It is not used here as the final classifier; instead, the Gini-based importance score it assigns to every feature is used purely to rank and filter the 25 candidate variables down to the six most informative ones before any classification tree is trained.
LMT — Logistic Model Tree	Combines a decision-tree structure with a logistic-regression model fitted at each node using the LogitBoost algorithm. Splits are chosen to maximise information gain, but instead of a simple majority-class leaf, each leaf outputs class probabilities from its local logistic model, blending statistical (regression-based) learning with tree-based inductive learning.
C4.5 Decision Tree	An extension of the classic ID3 algorithm that splits nodes using the Information Gain Ratio (IGR) rather than raw information gain, which corrects ID3's bias toward features with many distinct values. Growth stops once the sample count in a node falls below a threshold, and the tree is pruned afterwards using an error-based criterion.
C5.0 Decision Tree	A faster, more memory-efficient successor to C4.5 that produces smaller trees, tolerates a higher error threshold on the training set through pre-pruning, and automatically discards low-importance features during growth, generally yielding a more compact and slightly more accurate tree than C4.5 on the same data.
CART — Classification and Regression Tree	Produces strictly binary splits (exactly two child branches per internal node) chosen to minimise the Gini impurity index. CART supports cost-complexity pruning, which lets misclassification costs be factored directly into how aggressively the tree is trimmed — the variant ultimately selected as the article's best-performing model.

Dataset: The study uses a retrospective cohort of 2,470 COVID-19 patients admitted to two hospitals (Sina and Besat) in Hamadan, western Iran, between February 2020 and July 2021, of whom 1,853 (75.0%) recovered and 617 (25.0%) died. Twenty-five demographic, clinical, and laboratory variables were recorded per patient. Records with missing or outlier values were removed after confirming the missingness was completely at random, leaving a clean dataset that was then randomly split into training and test subsets ten separate times to reduce the effect of any single random split, with performance averaged across the ten repetitions. All analysis was carried out in Python 3.8 (scikit-learn) and R 4.1.2 (the train package), which is a fairly standard, reproducible tool stack for this kind of tabular clinical-ML study.

Feature Categories Recorded

Category	Examples of Recorded Variables	Count / Note
Demographic	Age, sex, marital status, location, smoking status	5 of 25 features
Clinical / Comorbidity	Hypertension, diabetes, cancer, cardiovascular disease, renal insufficiency, C.I.S., lung disease, hepatic failure, neurological disease	9 of 25 features
Laboratory Biomarkers	BUN, SGOT, ESR, PTT, platelets, sodium, LDH, CPK, blood sugar, PMN	11 of 25 features
Retained after RF selection	Sex, Age, BUN, PTT, SGOT, ESR (relative importance > 6%)	6 features used in final trees

Data preparation followed three explicit steps before any model was trained: (1) feature selection via Random Forest importance ranking, (2) comparison of the four candidate decision trees on ten repeated train/test splits, and (3) interpretation of the single best tree via its flowchart and ROC curve. This staged design deliberately separates 'which variables matter' from 'how to classify with them,' which is itself a useful methodological pattern for any inductive-learning pipeline built on tabular clinical data.

(b) Mapping to CO4 and Strength / Limitation

The methodology maps onto several CO4 sub-topics simultaneously: the CART, C4.5, C5.0, and LMT models are direct instances of Decision Tree algorithms; the process of learning classification rules from labelled training records is an example of Inductive Learning; and the LMT model, together with the accuracy/recall/F1/AUC evaluation framework, draws on Statistical Learning Methods for model comparison and decision-making support.

Methodological strength: the deliberate choice to keep the final predictive model as a single, readable decision tree — rather than the random forest itself — preserves clinical interpretability, letting physicians trace exactly which feature thresholds (e.g., BUN, age, SGOT) drive a given risk estimate, which is essential for trust and adoption in a medical decision-support setting.

Methodological limitation: the data come from only two hospitals in a single Iranian city, and the 75:25 class imbalance between recovered and deceased patients is not addressed through resampling, class weighting, or cost-sensitive learning, which can bias a tree toward the majority class and inflate apparent accuracy while leaving the minority (death) class under-served.

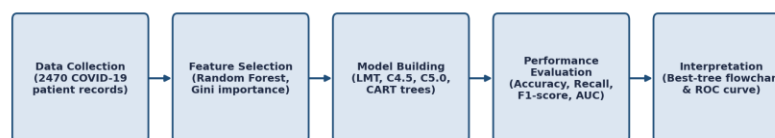
Additional Strengths Observed

- Reproducibility: the ten-times-repeated train/test split and averaging of Accuracy/Recall/F1-score across repetitions is a simple but effective way to reduce the influence of any one lucky or unlucky random split, which strengthens confidence in the reported numbers.
- Clinical relevance of chosen features: the six retained predictors (age, sex, BUN, PTT, SGOT, ESR) are all inexpensive, routinely collected values in most hospital laboratories, so the final model does not require any exotic or costly test to be applied at the bedside.
- Transparent reporting: the paper openly states its own limitations (single-region data, missing-data exclusion) rather than presenting the results as universally generalisable, which is good scientific practice for a clinical-ML paper.

Additional Limitations Observed

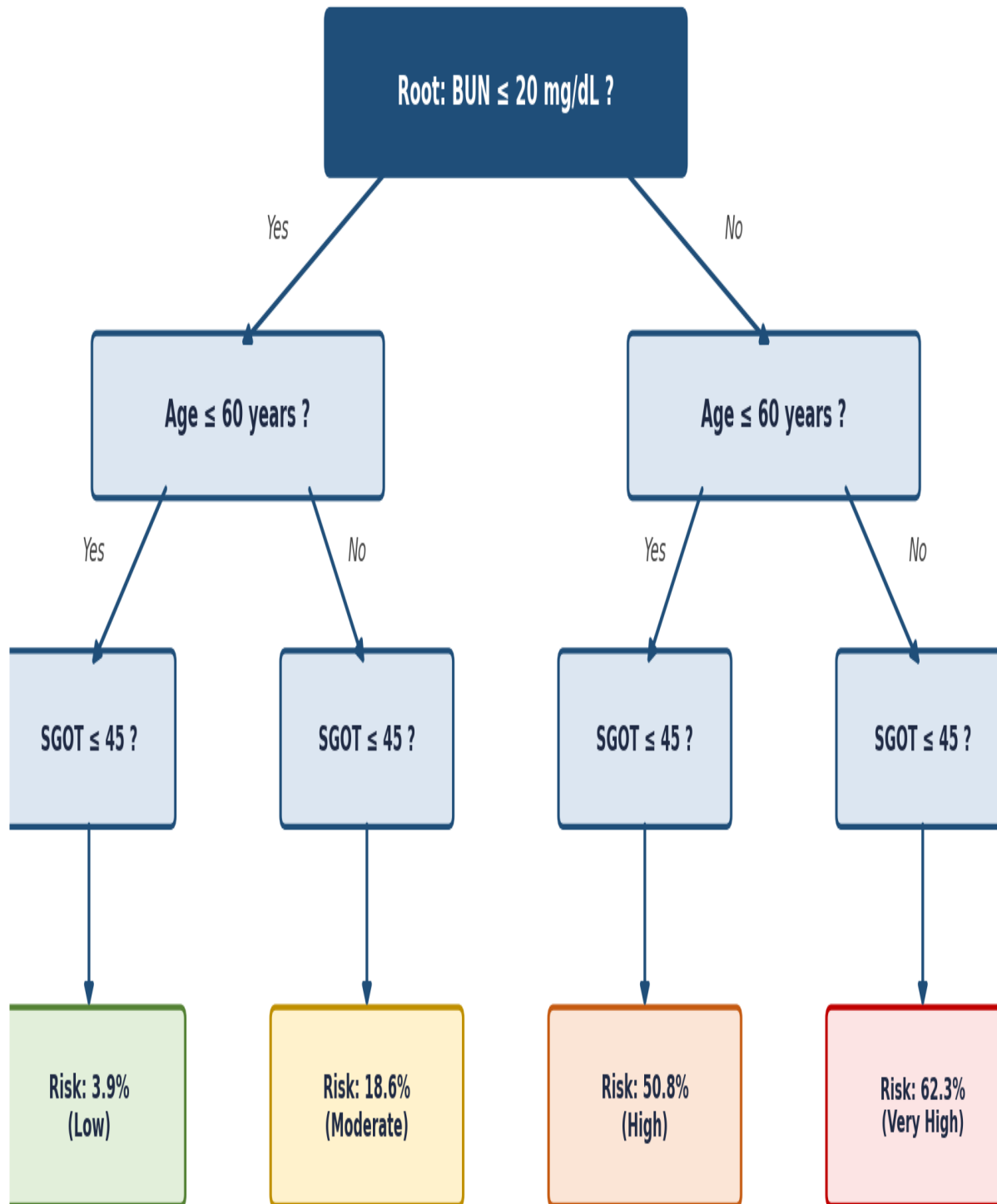
- No external validation cohort: all evaluation is performed on data drawn from the same two hospitals and the same outbreak period, so the reported accuracy may not transfer to a different population, healthcare system, or COVID-19 variant.
- Narrow performance gap treated as decisive: the four trees' F1-scores span only 85.73%–86.81%, a difference of about one percentage point, yet CART is described as clearly 'the best' model without any statistical test confirming the difference is significant rather than due to chance in the ten-fold split.
- No comparison against a non-tree baseline: the study benchmarks four decision-tree variants against each other but never against a simple logistic-regression baseline (independent of a tree) or an ensemble method such as gradient boosting, making it hard to judge how much the tree structure itself is adding over a simpler statistical model.

Methodology Overview



Illustrative CART Decision Path

The figure below is a simplified, illustrative reconstruction (not a verbatim reproduction) of how the article describes its selected CART tree branching on BUN, age, and SGOT to arrive at very different mortality-risk percentages for different patient profiles — this is the kind of leaf-level interpretability the authors highlight as the paper's main practical contribution.



Task 3: Results & Critical Evaluation

(a) Key Results and Comparative Table

Using the random-forest feature-ranking step, six variables — sex, age, blood urea nitrogen (BUN), partial thromboplastin time (PTT), SGOT, and ESR — were retained as the strongest mortality predictors (relative importance above 6%). Table 1 below lists the full importance tier reported in the article.

Feature	Category	Relative Importance
Age	Demographic	> 6% (top predictor)
Sex	Demographic	> 6%
BUN (Blood Urea Nitrogen)	Laboratory	> 6%
SGOT	Laboratory	> 6%
PTT	Laboratory	> 6%
ESR	Laboratory	> 6%
Hypertension, Diabetes, CVD, Marital status, Na, CPK, Plat	Demographic / Clinical / Laboratory	≈ 4.5%–6% (secondary tier)

All four decision-tree variants were then benchmarked on the training dataset; the CART model achieved the best combined accuracy and F1-score and was selected as the final interpretable model (Table 2).

Decision Tree Model	Accuracy (Training)	F1-Score (Training)	Rank (by F1-Score)
LMT	76.89%	85.73%	4th
C4.5	77.70%	86.08%	2nd
C5.0	76.89%	85.74%	3rd
CART (Best Model)	78.24%	86.81%	1st

The selected CART model was then evaluated on the held-out test set and on the total dataset, where it achieved an AUC of 79.5%, indicating moderate-to-good overall discriminative ability (Table 3).

Dataset Split	Accuracy	Recall	F1-Score
Test set	78.24%	95.50%	86.81%
Total dataset	78.34%	93.85%	86.67%

(b) Critical Evaluation of the Results

The reported metrics are broadly realistic for a single-center clinical dataset of this size, but they should be read with caution. Accuracy in the high-70% range together with an AUC of only 79.5% shows the model is far from a definitive diagnostic tool; recall for the training set is comparatively high (up to 95.5% on the test split), which may partly reflect the tree favouring the majority (recovered) outcome pattern rather than uniformly strong performance on both classes, since the 75:25 class imbalance is never explicitly corrected.

Potential Sources of Bias

- Selection bias: patients with any missing or outlier value were removed entirely, which could systematically exclude more severe, more complex, or more urgently treated cases rather than a random cross-section of admissions.
- Evaluation-protocol bias: the ten repeated random splits are not accompanied by confidence intervals or paired statistical-significance testing between the four tree models, whose F1-scores in fact differ by less than one and a half percentage points — too small a gap to declare a clear winner without a formal test.
- Geographic / institutional bias: both hospitals are in one Iranian province, so case-mix, testing protocols, and the dominant virus variant during the study window are not necessarily representative of other regions or later outbreak waves.
- Class-imbalance bias: with 75% of patients recovering, a tree can achieve deceptively high overall accuracy simply by leaning toward predicting recovery, which the paper does not explicitly rule out through a class-weighted or resampled comparison.

Additional Experiments That Would Strengthen the Findings

- External validation on an independent, multi-center cohort from a different country or health system to test whether the six selected predictors and the CART thresholds still hold.

- Reporting of confidence intervals or a paired statistical test (e.g., McNemar's test) across the four tree models rather than a single point-estimate comparison.
- An explicit class-imbalance treatment — SMOTE oversampling, class weighting, or cost-sensitive splitting — with a documented before/after comparison of Recall and F1-score for the minority (deceased) class specifically.
- Calibration analysis (e.g., a reliability diagram) of the predicted leaf-level risk percentages, to check whether a stated '62.3% risk' leaf genuinely corresponds to roughly 62% observed mortality among patients who land there.

Comparison with Related Studies

Placing the reviewed article's AUC of 79.5% alongside similar COVID-19 mortality-prediction studies cited in its own discussion section shows it sits toward the lower end of the reported range, which is a useful sanity check on how 'strong' the result really is (Table 4).

Study	Method	Reported AUC / Accuracy
Moslehi et al. (2022) — reviewed article	CART decision tree	AUC 79.5%; Accuracy 78.24%
Bhatia et al. (2021)	Decision tree (severity triage)	AUC 88.3%; Accuracy 79.74%
Yadaw et al. (2020)	ML on lab findings	AUC $\approx$ 91%
de Moraes Batista et al. (2020)	ML on ER admission data	AUC $\approx$ 85%

This comparison suggests the CART model's main advantage over the higher-AUC studies is not raw discriminative power but interpretability — a trade-off the article accepts deliberately, though it is worth naming explicitly rather than leaving implicit.

(c) Real-World Applicability

Because the final CART model needs only six routinely collected demographic and laboratory values, it is well suited to deployment as a lightweight clinical decision-support widget in an emergency department or triage dashboard, particularly in resource-limited settings during a surge, where it could help prioritise patients for closer monitoring or earlier intervention. Table 5 outlines several concrete deployment scenarios.

Emergency Triage Dashboard	Flag high-BUN, older, high-SGOT patients on admission for closer monitoring using only six routinely available values.
Resource-Limited / Field Hospitals	Low compute cost (a single tree) makes the model runnable on basic hardware where cloud-based deep-learning tools are impractical.
Clinical Decision-Support (CDS) Plugin	Embed the flowchart logic inside an existing hospital information system so risk bands appear automatically alongside lab results.
Training Tool for Junior Clinicians	The transparent branch structure doubles as a teaching aid for what combinations of findings historically correlated with worse outcomes.

Scaling the approach to broader industry or multi-hospital use, however, faces several challenges: data availability and standardisation across different electronic health record systems and laboratories (unit conventions, reference ranges); the need for prospective, multi-site validation before any clinical deployment; computational needs are modest (a single decision tree is cheap to run), so compute is not the bottleneck — governance is. Ethical and regulatory issues include patient-data privacy and consent for secondary use, the risk of automation bias if clinicians over-trust a single risk percentage, and the need for periodic retraining as new COVID-19 variants or treatment protocols shift the underlying risk relationships captured by the tree.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights Connected to CO4

Insight 1 — Inductive Learning in Practice

The CART, C4.5, C5.0, and LMT models all learn their classification rules purely from labelled historical patient records, with no hand-coded medical rules supplied in advance. This is a direct, real-world illustration of the CO4 concept of Inductive Learning: generalising a decision rule from specific training examples so that it can classify new, unseen patients. What stood out to me is that the 'rule' the tree induces (e.g., $BUN \leq 20$ and $age \leq 60$ and $SGOT \leq 45 \rightarrow$ low risk) is not something a domain expert typed in; it emerged purely from statistical regularities in 2,470 patient records, which is the essence of inductive reasoning applied to a real clinical dataset rather than a toy classroom example.

Insight 2 — Ensembles and Single Trees Serve Different Purposes

The article uses a Random Forest (ensemble of statistical/decision-tree learners) purely for feature ranking and then deliberately falls back to a single interpretable tree for the final classifier. This deepened my understanding that CO4's Statistical Learning and Decision Tree topics are complementary tools chosen for different jobs, not competing options where one is simply 'better.' An ensemble is well suited to robustly ranking which variables matter because averaging over 1,000 trees smooths out noise, while a single tree is far better suited to the deployment goal of transparent, explainable decision-making — the study's design choice makes that trade-off explicit rather than defaulting to whichever model scores highest.

Insight 3 — Evaluation Metrics Must Match the Decision Context

Relying on accuracy alone would have hidden the class-imbalance issue in this 75:25 dataset; reporting Recall, F1-score, and AUC-ROC together, as the CO4 syllabus emphasises for decision-making models, gives a far more honest picture of how a model will behave when the minority (mortality) outcome is the one that matters most clinically. This reframed, for me, why a course on decision-making models spends so much time on evaluation metrics rather than treating model-fitting as the whole job — in a healthcare setting, the cost of a false negative (missing a high-risk patient) and a false positive (over-flagging a low-risk patient) are simply not symmetric, and the choice of metric has to reflect that asymmetry.

(b) Proposed Extension / Novel Experiment

Proposed enhancement: extend the study by (i) pooling data from multiple hospitals across different provinces or countries (e.g., using a publicly available multi-center critical-care dataset such as MIMIC-IV) to test generalisability, (ii) applying an explicit class-imbalance correction such as cost-sensitive CART splitting or SMOTE before retraining, and (iii) adding a simple reinforcement-learning layer on top of the tree that updates a patient's triage priority dynamically as new lab results arrive during their stay, rather than issuing a single static prediction at admission — directly bridging the Decision Tree and Reinforcement Learning strands of CO4 in one applied system.

Proposed Study Timeline

Weeks 1–3	Secure ethics approval / data-use agreements; assemble the multi-center dataset (e.g., pooled hospital records or a public critical-care dataset such as MIMIC-IV); harmonise units and reference ranges across sites.
Weeks 4–6	Re-run the Random-Forest feature-selection step on the pooled data; verify whether the same six variables (age, sex, BUN, PTT, SGOT, ESR) remain the top predictors across sites, or whether new region-specific predictors emerge.
Weeks 7–9	Retrain LMT, C4.5, C5.0, and CART with an explicit class-imbalance correction (cost-sensitive splitting and SMOTE, compared side by side); benchmark against the original single-center results.
Weeks 10–12	Prototype the reinforcement-learning re-prioritisation layer on top of the best tree, using simulated sequential lab-result updates; evaluate against the static, admission-time-only baseline.
Weeks 13–14	External validation on a held-out site not used in training; write up calibration analysis and comparative results.

Feasibility and Expected Impact

Feasibility: multi-center, de-identified COVID-19 and general critical-care datasets are already publicly available for research use, and the class-imbalance techniques proposed are standard, low-cost additions to the existing scikit-learn/R pipeline the original authors already used, so no new infrastructure is required. The main practical hurdle is data-governance and ethics approval for pooling records across institutions, not a lack of methods or compute.

Expected impact: a model that is both more generalisable across health systems (through multi-center training and explicit bias correction) and more clinically useful (through dynamic, reinforcement-learning-based re-prioritisation rather than a single admission-time risk score), while retaining the interpretability that made the original CART model attractive to clinicians. If successful, this extension would demonstrate that Decision Trees, Statistical Learning, and Reinforcement Learning — the three pillars of CO4 — can be combined in one coherent clinical pipeline rather than treated as three separate, unrelated topics.

Glossary of Key Terms

The following short glossary collects the CO4-relevant technical terms used throughout this review, for quick reference during evaluation.

Inductive Learning	The process of learning a general rule or hypothesis from specific labelled examples, so that the rule generalises correctly to new, unseen cases.
Decision Tree	A tree-structured classification model where internal nodes test a feature against a threshold and leaves assign a class label or probability.
CART	Classification and Regression Trees — produces strictly binary splits chosen to minimise Gini impurity; supports cost-complexity pruning.
C4.5 / C5.0	Successive decision-tree algorithms that split nodes using an information-gain-ratio criterion; C5.0 is a faster, more compact refinement of C4.5.
LMT (Logistic Model Tree)	A hybrid model that fits a logistic-regression model at each tree node instead of a simple majority-class leaf label.
Random Forest	An ensemble of many decision trees trained on bootstrap samples; used here purely to rank feature importance, not as the final classifier.
Accuracy	The proportion of all predictions (both classes) that are correct: $(TP + TN) / (TP + TN + FP + FN)$.
Recall (Sensitivity)	The proportion of actual positive cases (e.g., deceased patients) that the model correctly identifies: $TP / (TP + FN)$.
F1-score	The harmonic mean of Precision and Recall, balancing both into a single number that penalises models which favour one at the expense of the other.
AUC-ROC	Area Under the Receiver Operating Characteristic curve; summarises a model's ability to distinguish between classes across all decision thresholds, from 0.5 (no better than chance) to 1.0 (perfect).
Reinforcement Learning	A learning paradigm in which an agent learns a policy by taking actions in an environment and receiving reward feedback, refining its behaviour over repeated interactions.

Reference List

Primary article reviewed:

- Moslehi, S., Rabiei, N., Soltanian, A. R., & Mamani, M. (2022). Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran. *BMC Medical Informatics and Decision Making*, 22, Article 192. <https://doi.org/10.1186/s12911-022-01939-x>

Comparative studies cited in Task 3(b), as referenced within the primary article's own discussion section:

- Bhatia, S., Makhija, Y., Jayaswal, S., Singh, S., & Gupta, I. (2021). Severity and mortality prediction models to triage Indian COVID-19 patients. *arXiv:2109.02485*.
- Yadaw, A. S., Li, Y.-C., Bose, S., Iyengar, R., Bunyavanich, S., & Pandey, G. (2020). Clinical predictors of COVID-19 mortality. *medRxiv*.
- de Moraes Batista, A. F., Miraglia, J. L., Donato, T. H. R., & Chiavegatto Filho, A. D. P. (2020). COVID-19 diagnosis prediction in emergency care patients: A machine learning approach. *medRxiv*.

Rubric Reference (for Evaluation)

Criteria	Weight	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Clear, in-depth understanding of purpose, concepts, and arguments	Good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Concise, accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Insightful critique with strong reasoning and evaluation	Reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand